

TrES-3b

R-2641

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160603 · created 2026-08-29T01:24:01+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457542.8076 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.017
Transit depth (Rp/Rs)^2	3.87 +/- 0.65 [%]
Orbital Inclination (inc)	81.93 +/- 0.78
Airmass coefficient 1 (a1)	0.993 +/- 0.011
Airmass coefficient 2 (a2)	0.007 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#3 - [522, 439]
Optimal Aperture	4.03
Optimal Annulus	9.65
Transit Duration (day)	0.0587 +/- 0.0072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.017 vs 0.16309 (deviation 1.99 σ)
Duration fit vs published	0.0587 d vs 0.0567 d (deviation 0.28 σ)
Mid-time O-C	-4.0 min
Depth SNR	11.6
Residual scatter	1.08 %

Light curve

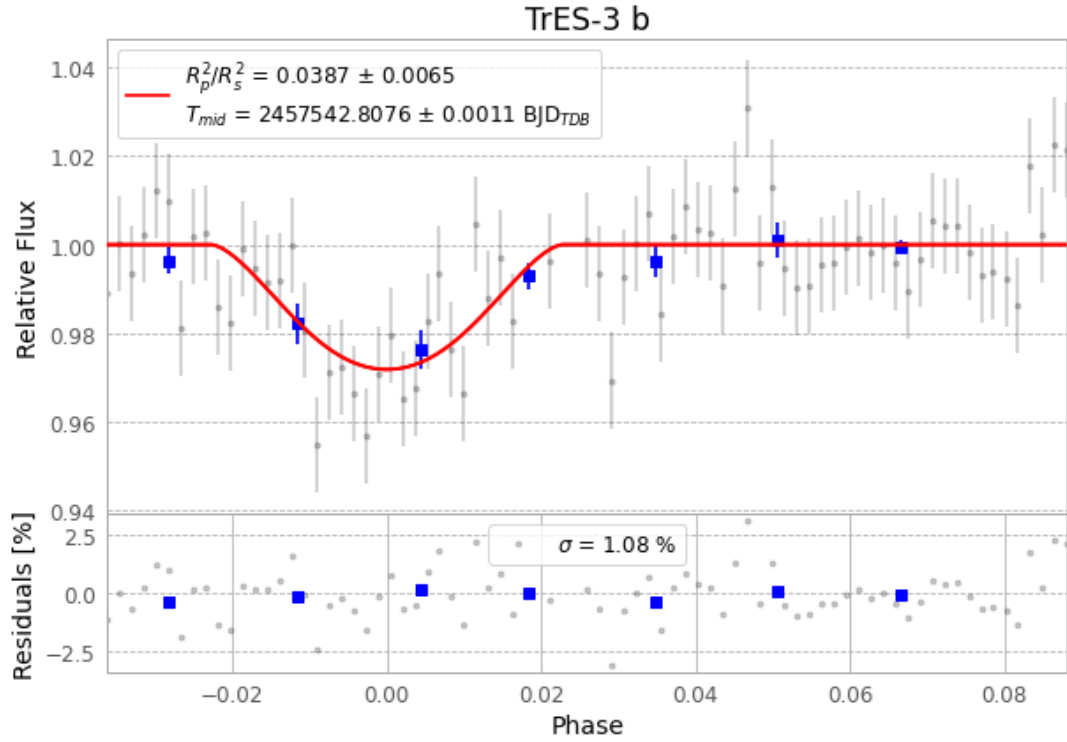


Figure 1 — FinalLightCurve_TrES-3 b_2016-06-02.png (SHA-256 2dc34811584b6d28...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 36b3bf885c9e4a37...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

75 FITS frames (49 MB), TRES-3160603060912.FITS → TRES-3160603100313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160603.log (604b09eb6907...), FinalLightCurve_TrES-3 b_2016-06-02.png (2dc34811584b...), FinalLightCurve_TrES-3 b_2016-06-02.csv (04da15d95f8c...)

Compute resources

Wall time 123.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 01:24Z.